MISSINGNESS-RELIABLE REGIONAL GRAPH FORECASTING FOR SPATIAL TIME SERIES

Author Name(s)

Author Affiliation(s)

ABSTRACT

Missing observations in spatial time series not only shorten temporal context but also make learned sensor relations unreliable. We propose MR-MSGF, a direct forecasting model that predicts from incomplete histories without a separately supervised imputation stage. MR-MSGF builds a window-specific relation graph whose learned similarities are regularized by pairwise co-observation, fuses it with physical topology, and coarsens the fused relation into physically connected regions. A lightweight reliability gate then controls when regional context should influence multi-scale prediction. On METR-LA, MR-MSGF yields consistent, paired-significant MAE reductions under contiguous and point-wise missingness. AQI36 native missingness and Block-T 70% RMSE further characterize the mechanism as a missingness-aware bias rather than a cross-domain performance guarantee.

***Index Terms*—** Multivariate time series forecasting, missing observations, spatiotemporal graph learning, graph coarsening

1. INTRODUCTION

Traffic, environmental, and industrial forecasting systems depend on temporal context and inter-sensor relations. Spatiotemporal graph models represent these relations with physical topology or learned adjacency [1, 2], while hybrid designs combine physical and data-driven dependencies [3]. Both relation types are used to propagate information across sensors. Under non-synchronous outages, two equally similar node representations can rest on very different amounts of joint observation. Missingness therefore affects not only node histories, but also whether an inferred edge has sufficient evidence for message passing.

Missing-data research primarily addresses the node-history problem. Imputation reconstructs values through bidirectional recurrent models [4], temporal attention for irregular samples [5], or graph propagation [6, 7]; direct forecasting instead encodes values and masks jointly [8, 9]. These routes make incomplete histories usable for prediction. Once their features enter graph learning, however, node availability alone does not state how much jointly observed history supports a candidate node pair.

Pairwise evidence is distinct from a node-wise missingness indicator. A large adaptive similarity can be estimated from only a few coincident observations, or from a long jointly observed history; its numerical strength is the same type of quantity, but its support is not. The distinction matters before top-k selection commits candidate relations to subsequent graph propagation.

Relation modeling under irregular observation has also been studied beyond forecasting. RAINDROP learns sample-specific sensor graphs from irregular observations for classification [10], while TGNN4I handles partial node observations and irregular times through continuous-time graph states [11]. For missing-data forecasting, BiTGraph injects missing-pattern biases into temporal and graph propagation [12]; GinAR combines predefined and adaptive graphs under variable missingness [13]; and GCN-M studies graph learning with incomplete traffic observations [14]. These methods model missingness together with node representations or graph construction, but do not explicitly separate a learned relation's strength from the amount of pairwise evidence supporting it.

Multi-scale models further aggregate context across time and space [15, 16, 17]. A physical topology supplies a stable locality prior that does not depend on the current missing window, whereas a learned graph can capture window-specific functional dependence. They are complementary, but neither source alone determines whether a coarse regional message is useful for a particular node. A connected region can still have sparse coverage, long outages, or a node representation far from its regional center. Fusing relations before physically connected regionalization retains both forms of evidence; conditioning the resulting regional context on such node, edge, and regional cues avoids treating coarse aggregation as universally beneficial.

This leaves a specific question: given incomplete history, how should a model decide whether a learned edge and the regional message induced by it are trustworthy? We study *relation reliability* as complementary to historical reconstruction. The goal is not to recover a ground-truth graph or calibrated uncertainty, but to condition forecasting messages on available evidence. MR-MSGF regularizes learned relations with co-observation, fuses them with physical topology before connected regionalization, and gates regional context with node, edge, and regional reliability evidence. Our

contributions are:

- We quantify *relation reliability* by regularizing learned edges with pairwise co-observation before sparse selection.
- We fuse observation-supported and physical graphs, regionalize the fusion, and use reliability evidence to gate regional messages and multi-scale readout.
- Paired METR-LA tests support this bias, while AQI36 and extreme outages define its scope.

2. METHOD

2.1. Problem Setup and Backbone

Let $X \in \mathbb{R}^{T\times N\times C}$ be a history window, $M \in \{0,1\}^{T\times N\times C}$ its observation mask, and $A^{\mathrm{phy}} \in \mathbb{R}^{N\times N}$ the physical adjacency. The model predicts $Y \in \mathbb{R}^{H\times N\times C_y}$ from $X^{\mathrm{obs}} = M \odot X$, M, and A^{phy}. Graphs and reliability statistics are computed per window; we omit the batch index. With channel-averaged mask $\bar{M}_{t,i} = C^{-1}\sum_c M_{t,i,c}$,

$$r_i = \frac{1}{T}\sum_t \bar{M}_{t,i}, \qquad q_{ij} = \frac{1}{T}\sum_t \bar{M}_{t,i}\bar{M}_{t,j}. \tag{1}$$

Here, r_i denotes availability and q_{ij} co-observation; all graphs are row-normalized. The retained encoder, temporal hierarchy, and multi-scale readout of [15] yield $H_\ell \in \mathbb{R}^{N\times d}$ and $B_{s,\ell}$. Here ℓ indexes temporal levels, $s = 0$ is the finest spatial level, and $s > 0$ are original coarser levels. Our branch modifies only $B_{0,\ell}$.

Fig. 1 outlines MR-MSGF: co-observation guides fine relations, whose regional context is reliability-gated for prediction.

2.2. Observation-Supported Relation Fusion

The encoder produces history features $E \in \mathbb{R}^{T\times N\times d}$. We obtain a window-specific node representation by masked pooling:

$$u_i = \frac{\sum_t \bar{M}_{t,i}E_{t,i}}{\sum_t \bar{M}_{t,i} + \epsilon}. \tag{2}$$

The observation-supported score combines learned similarity and co-observation:

$$s_{ij} = \frac{(W_q u_i)^\top (W_k u_j)}{\sqrt{d}}, \tag{3}$$

$$\tilde{s}_{ij} = s_{ij} + \mathrm{softplus}(\beta)\log(q_{ij} + \epsilon). \tag{4}$$

Here β is learnable. Retaining top-k scores per node, applying row softmax, adding self-loops, and row-normalizing yields directed sparse A^{obs} ($k = 8$). The logarithmic term softly discourages edges with little joint observation before sparsification.

Let $\bar{A}^{\mathrm{phy}}$ denote the physical graph after adding self-loops and applying row normalization. A learned edge gate then forms the fine relation:

$$g_{ij} = \mathrm{sigmoid}\Big(\psi_g([r_i, r_j, q_{ij}, \bar{A}^{\mathrm{phy}}_{ij}, A^{\mathrm{obs}}_{ij}])\Big), \tag{5}$$

$$A^{\mathrm{fine}} = \mathrm{RowNorm}\big(g \odot \bar{A}^{\mathrm{phy}} + (1-g)\odot A^{\mathrm{obs}}\big). \tag{6}$$

Thus, both physical structure and observation-supported relations participate in end-to-end learned relation fusion. The gate is intentionally edge-specific rather than a global mixing coefficient, allowing physical and observation-supported evidence to receive different weights for different edges. In implementation, ψ_g is a two-layer $5 \to 16 \to 1$ MLP with GELU activation.

2.3. Reliability-Controlled Regional Context

We construct a fixed hard assignment $S \in \{0,1\}^{N\times K}$ from the physical graph, with one region per node. After symmetrizing the adjacency and removing self-loops, we seed by remaining physical degree and greedily add the unassigned node with the largest positive affinity to a region member until no such node remains or size $m = 8$ is reached. Thus regions are physically connected and do not absorb zero-affinity nodes. With $n_k = \sum_i S_{ik}$,

$$Z^{\mathrm{fine}}_\ell = \mathrm{ELU}(A^{\mathrm{fine}}H_\ell W_f + H_\ell W_s), \tag{7}$$

$$H^{\mathrm{reg}}_{\ell,k} = \frac{1}{n_k}\sum_i S_{ik}Z^{\mathrm{fine}}_{\ell,i}, \tag{8}$$

$$A^{\mathrm{coarse}} = \mathrm{RowNorm}(S^\top A^{\mathrm{fine}} S). \tag{9}$$

Regional propagation and lifting are

$$Z^{\mathrm{reg}}_\ell = \mathrm{ELU}(A^{\mathrm{coarse}}H^{\mathrm{reg}}_\ell W_c + H^{\mathrm{reg}}_\ell W_r), \tag{10}$$

$$Z^{\mathrm{coarse}}_\ell = SZ^{\mathrm{reg}}_\ell. \tag{11}$$

Coarsening after fine-scale fusion lets regional aggregation operate on a graph containing both physical structure and observation-supported relations; coarsening only the physical or observation graph cannot retain both sources of evidence.

For node i, let $\bar{q}_i = \sum_j \bar{A}^{\mathrm{phy}}_{ij}q_{ij}$ be physical-support-weighted co-observation, $\rho_i = \sum_k S_{ik}n_k^{-1}\sum_j S_{jk}r_j$ be regional coverage, and b_i be the longest missing block in the current history window divided by T. Let $k(i)$ denote the region of node i, $c_{i,\ell} = H^{\mathrm{reg}}_{\ell,k(i)}$, and

$$\zeta_{i,\ell} = \sqrt{\max\Big(d^{-1}\|Z^{\mathrm{fine}}_{\ell,i} - c_{i,\ell}\|_2^2, \epsilon\Big)}, \tag{12}$$

$$\delta_{i,\ell} = \frac{\zeta_{i,\ell}}{N^{-1}\sum_j \zeta_{j,\ell} + \epsilon}. \tag{13}$$

Here $\delta_{i,\ell}$ is normalized representation dispersion from the re-

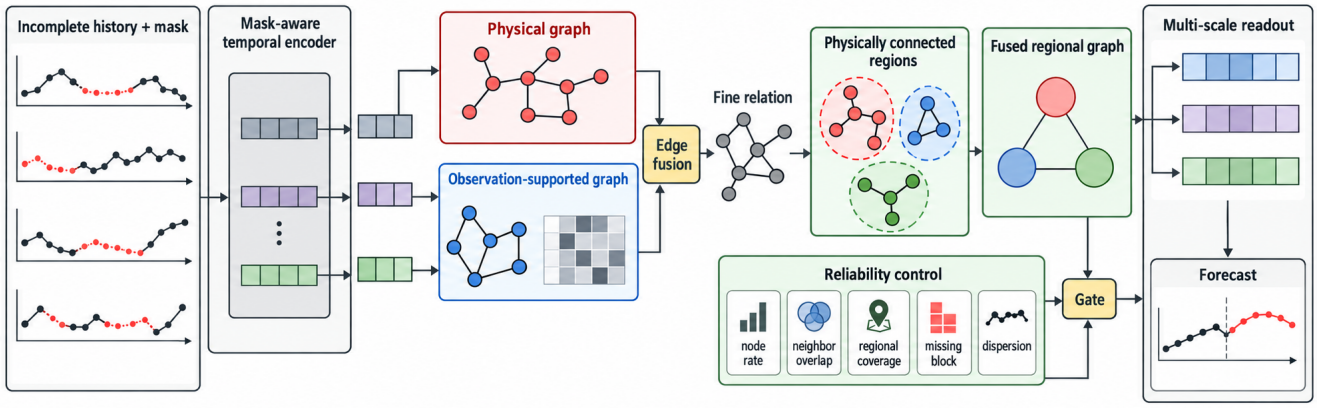


Fig. 1. Overview of MR-MSGF. A mask-aware encoder forms temporal states from incomplete history. A co-observation-supported graph is fused with the physical graph before physically connected regional coarsening. Reliability evidence regulates regional residual injection and multi-scale readout.

gional center. The regional gate is

$$\alpha_{i,\ell} = \text{sigmoid}\left(\psi_\alpha([r_i, \bar{q}_i, \rho_i, b_i, \delta_{i,\ell}]) \right. \tag{14}$$

$$\left. + \gamma^\top [1 - r_i, 1 - \bar{q}_i, \rho_i - 1, b_i, -\delta_{i,\ell}]\right). \tag{15}$$

Here ψ_α uses the same two-layer MLP structure as the edge gate, and γ is a learnable scaling vector. The second term supplies missingness patterns and regional consistency as an additional residual signal. Since γ is not sign-constrained, we do not interpret this term as a monotonic prior. Let $\boldsymbol{\alpha}_\ell = [\alpha_{1,\ell}, \ldots, \alpha_{N,\ell}]^\top \in \mathbb{R}^N$, broadcast along the feature dimension. The finest representation becomes

$$\widetilde{B}_{0,\ell} = Z_\ell^{\text{fine}} + \boldsymbol{\alpha}_\ell \odot \left(Z_\ell^{\text{coarse}} W_p\right), \tag{16}$$

while $\widetilde{B}_{s,\ell} = B_{s,\ell}$ for $s > 0$. We further bias multi-scale readout by reliability evidence. Let $R_i = [r_i, \bar{q}_i, \rho_i, b_i]^\top$. The score for prediction step h is

$$e_{h,s,\ell,i} = e^{\text{repr}}_{h,s,\ell,i} + w_{s,\ell}^\top R_i. \tag{17}$$

For each node and prediction step, a softmax over (s, ℓ) yields $\pi_{h,s,\ell,i}$, and $F_{h,i} = \sum_{s,\ell} \pi_{h,s,\ell,i} \widetilde{B}_{s,\ell,i}$ is mapped to $\hat{Y}_{h,i}$ by the prediction head. Thus, reliability statistics bias scale selection rather than replacing representation content. Training minimizes masked forecasting MAE,

$$\mathcal{L}_{\text{pred}} = \frac{\sum M^Y \odot |\hat{Y} - Y|}{\sum M^Y}, \tag{18}$$

where M^Y excludes unavailable future targets. We do not use reconstruction, graph, or clustering auxiliary losses. The gate is not interpreted as calibrated uncertainty. It is a forecasting-oriented control variable that decides whether regional context should be added for a node at a temporal level. This design keeps the added branch lightweight while making regional messages conditional on missingness evidence rather than unconditional smoothing.

3. EXPERIMENTS

3.1. Settings

We evaluate METR-LA traffic speeds and AQI36 air quality, forecasting 12 steps from 24 histories. METR-LA Point and Block-T masks affect only history; valid, unmasked raw targets are evaluated. Point simulates independent loss and Block-T contiguous outages. AQI36 retains its native validity mask with optional Point masks. Conditions use five paired mask seeds and shared input masks. Training uses only Eq. (18). We use $k = m = 8$, initialize $\beta = 1$ and γ to ones, and obtain $K = 31$ and $K = 5$ connected regions for METR-LA and AQI36. HD-TTS-AMP is the main reference; BiTGraph uses the common data, mask, and evaluation interface.

3.2. Main Results

Table 1 reports METR-LA results. MR-MSGF lowers MAE over HD-TTS-AMP in all tested Block-T and Point settings. It also improves RMSE under Block-T 30%/50% and Point 30%/50%. At Block-T 70%, MR-MSGF still improves MAE but has slightly worse RMSE, indicating that the benefit is not cross-metric consistent under extreme missingness. Across

Table 1. METR-LA results over five paired mask seeds. Lower is better; the best result within each condition is in bold.

Mask	Rate	Method	MAE	RMSE
Block-T	30%	HD-TTS-AMP	3.2281 ± 0.0108	6.6036 ± 0.0126
		BiTGraph	3.8232 ± 0.0110	7.8839 ± 0.0373
		MR-MSGF	$\mathbf{3.1997 \pm 0.0159}$	$\mathbf{6.5840 \pm 0.0485}$
	50%	HD-TTS-AMP	3.3546 ± 0.0126	6.8780 ± 0.0215
		BiTGraph	3.9729 ± 0.0180	8.1905 ± 0.0458
		MR-MSGF	$\mathbf{3.3079 \pm 0.0195}$	$\mathbf{6.8154 \pm 0.0554}$
	70%	HD-TTS-AMP	3.4650 ± 0.0103	$\mathbf{7.0927 \pm 0.0421}$
		BiTGraph	4.1363 ± 0.0132	8.4812 ± 0.0306
		MR-MSGF	$\mathbf{3.4458 \pm 0.0102}$	7.1088 ± 0.0384
Point	30%	HD-TTS-AMP	3.1048 ± 0.0074	6.3592 ± 0.0154
		BiTGraph	3.6374 ± 0.0124	7.5260 ± 0.0377
		MR-MSGF	$\mathbf{3.0764 \pm 0.0118}$	$\mathbf{6.3091 \pm 0.0447}$
	50%	HD-TTS-AMP	3.1598 ± 0.0112	6.4755 ± 0.0215
		BiTGraph	3.6938 ± 0.0049	7.6577 ± 0.0289
		MR-MSGF	$\mathbf{3.1409 \pm 0.0068}$	$\mathbf{6.4564 \pm 0.0289}$

Table 2. AQI36 cross-domain results over five seeds.

Condition	Method	MAE	RMSE
Native	HD-TTS-AMP	$\mathbf{30.4839 \pm 0.5849}$	$\mathbf{49.3332 \pm 1.4069}$
	MR-MSGF	30.7269 ± 0.2392	49.8879 ± 0.9287
Native+P25	HD-TTS-AMP	31.2034 ± 0.6894	50.3712 ± 1.0346
	MR-MSGF	$\mathbf{31.1169 \pm 0.4161}$	$\mathbf{50.3325 \pm 0.7815}$
Native+P50	HD-TTS-AMP	31.7359 ± 0.4956	$\mathbf{50.4029 \pm 1.2035}$
	MR-MSGF	$\mathbf{31.6451 \pm 0.7093}$	50.5270 ± 1.3859

all five conditions, MR-MSGF wins all five paired MAE runs and two-sided paired t-tests give $p \leq 0.0379$; we make no analogous across-condition significance claim for RMSE.

3.3. Cross-Domain and Horizon Analysis

AQI36 provides a cross-domain stress test with native missingness. Table 2 shows that MR-MSGF is worse than HD-TTS-AMP under native missingness, improves both metrics with an additional 25% Point mask, and lowers MAE but raises RMSE with an additional 50% Point mask. The heterogeneous native missingness need not make the current reliability statistics align with transferable relations; this is a hypothesis suggested by the result, not a causal explanation. We therefore treat AQI36 as a boundary condition rather than claiming dataset-independent gains.

We also decompose METR-LA Block-T checkpoints by prediction horizon. At 50% missingness, MR-MSGF lowers both MAE and RMSE at horizons 3, 6, and 12. At 70% missingness, it slightly lowers MAE at all three horizons but raises RMSE at each horizon, matching the aggregate result in Table 1. The extreme-missingness gain is therefore concentrated on average error rather than large-error control.

Table 3. METR-LA Block-T horizon decomposition.

Rate	Horizon	HD-TTS-AMP MAE/RMSE	**MR-MSGF** MAE/RMSE
50%	3	$3.1319/6.2557$	$\mathbf{3.0750}/\mathbf{6.1586}$
50%	6	$3.3787/6.9274$	$\mathbf{3.3333}/\mathbf{6.8715}$
50%	12	$3.6532/7.6045$	$\mathbf{3.6195}/\mathbf{7.5840}$
70%	3	$3.2773/\mathbf{6.5703}$	$\mathbf{3.2520}/6.5811$
70%	6	$3.4777/\mathbf{7.1235}$	$\mathbf{3.4667}/7.1568$
70%	12	$3.7302/\mathbf{7.7343}$	$\mathbf{3.7186}/7.7618$

Table 4. METR-LA Block-T 50% ablation over five paired seeds.

Variant	MAE	RMSE
HD-TTS-AMP	3.3546 ± 0.0126	6.8780 ± 0.0215
Physical only	3.3292 ± 0.0118	6.8605 ± 0.0490
Observation only	3.3348 ± 0.0161	6.8874 ± 0.0587
Fused graph	3.3176 ± 0.0111	6.8288 ± 0.0455
No q_{ij} penalty	3.3149 ± 0.0200	6.8327 ± 0.0547
No rel. gate	3.3136 ± 0.0105	6.8395 ± 0.0420
No readout bias	3.3298 ± 0.0249	6.8512 ± 0.0375
Full **MR-MSGF**	$\mathbf{3.3079 \pm 0.0195}$	$\mathbf{6.8154 \pm 0.0554}$

3.4. Ablation and Diagnostics

Table 4 isolates the Block-T 50% mechanisms. The full model is best on both metrics. Removing the co-observation score penalty, the reliability gate, or the readout bias all hurts performance, and using only one fine relation source is weaker than the fused graph. This supports the central design choice: co-observation-guided relation fusion and reliability-controlled regional context are complementary.

4. CONCLUSION

MR-MSGF forecasts incomplete spatial histories by making learned graph relations conditional on co-observation evidence, then fusing them with physical structure and gating regional context. The paired METR-LA results support relation reliability as a useful missingness-aware inductive bias. The mixed RMSE outcome at Block-T 70% and AQI36 native-missingness result delimit this claim: the mechanism is not a cross-domain or cross-metric performance guarantee.

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